

Real-time OEE visualisation for downtime detection

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Abstract—Unknown and unplanned downtime events during production cause significant disruption and loss of productivity. Investigating, identifying and addressing such events is a pressing need. The primary objective of this study is to examine downtime, performance loss, and quality control in the manufacturing process. Specifically, we propose a solution that provides real-time data processing and visualization of the factory floor. This solution was implemented for a major food manufacturer based in New Zealand. The company provided historical data covering over six years of operation and access to real-time data through their Industrial Internet of Things (IIoT) systems executing on Programmable Logic Controllers (PLCs). Our solution is an Overall Equipment Effectiveness (OEE) standardized Supervisory Control and Data Acquisition (SCADA) system that visualizes the manufacturing process in real-time. Analysis of the data collected during this research shows that by implementing the OEE and employing shift adjustment, there was a significant increase in production output. OEE can help improve manufacturing performance by pinpointing the root of the loss of performance in all areas monitored.

Keywords—Data analysis, Big data, OEE, Industry 4.0, SCADA, Real-time OEE

I. INTRODUCTION

Industry 4.0 has driven the performance and efficiency of manufacturing by automation and data exchange. One of the challenges in Industry 4.0 is to predict and prevent abnormal behaviors in the manufacturing process. Standardized Supervisory Control and Data Acquisition (SCADA) has played an essential role in the migration to Industry 4.0, where the processing data is connected through the Industrial Internet of Things (IIoT). In this case, real-time data monitoring is becoming extremely important for manufacturing control regarding quality and performance. Overall Equipment Effectiveness (OEE) is the gold standard to measure the manufacturing process [1]. Real-time OEE enables manufacturers to have a real-time analytics dashboard, an intelligent alert system and provides transparency of the production line which produces higher levels of production and less waste [2].

Previous research has demonstrated that the use of the SCADA system is to control a network of Programmable Logical controllers (PLC) and Remote Terminal Units (RTU). These are able to measure the factory's performance and gain insights into the machinery status, SCADA is commonly used in the food industry, water systems and wind turbine plants [3] [4] [5].

However, there is limited research on implemented standardized measures in the industrial real-time data

monitoring application. The use of data analytics to visualize technical engineering language to non-technical supervisors is still a challenge in this field. It is critical to make interactive dashboards/reports for management to view in real-time to achieve better business decisions. Hence, transforming the production line in the respect of OEE into Key Performance Indicator (KPI) is the key to making better decisions [6].

For the case study company Leaderbrand, reducing downtime is a significant goal. Factors contributing to downtime include the lack of standardization, lack of skilled human-resources, and a significant personnel turnover. It is recommended they use the standardization process to secure gain in their productivity and provide real-time status updates on the product quality. To counter those mentioned downtime issues, this project aims to provide interactive dashboards that drive the production line's performance, reduce the downtime frequency and therefore increasing production capacity.

In this research project, we used the SCADA system to design and integrate OEE into live dashboards, in addition, we enabled production analysis in the system that provides real-time traceability, visibility, and interoperability to illustrate the execution against the plan [2]. Therefore, management teams and engineers can have better insights and have the ability to enhance assets' up-time.

A scoping study was conducted to map this project's primary functionalities which provides insights into the manufacturing process. We outlined our studies via three research questions as followed.

- 1) What are the mostly common real-time data monitoring techniques and the aspects of OEE measuring employed in the manufacturing process?
- 2) What are the research gaps and challenges in real-time OEE monitoring that are highlighted in the state of the art solution?
- 3) How can the enterprise achieve better business decisions from OEE and real-time data monitoring in relation to downtime events?

II. BACKGROUND

Currently, LeaderBrand¹ faces difficulties with unplanned downtime in their operations due to multiple external factors such as machine breakdown, personnel turnover and lack of raw material. It has caused lost production, lost capacity, and increased inventory expense. It is a major goal for Leaderbrand

¹<https://www.leaderbrand.co.nz/>

to investigate and provide a solution to standardize their manufacturing process, and to secure consistent gain in their productivity. Therefore, they wish to identify and target weakness in their production line, and hence then being able to allocate resources efficiently, reducing human errors and overall improving production output.

OEE has been widely used in the manufacturing industry, especially in the manufacturing maintenance process [1]. It is the gold standard to measure how well a manufacturing operation is utilized compared to its full potential. The objective of OEE is to identify a single asset or a single process line, and OEE has three major elements: 1. Availability indicates the uptime of the machine, 2. Performance presents the machine's production rate against the theoretical rate. 3. Quality shows the quality control of the products [1] [7] [8]. In our scenario for quality metrics, good pieces are the product's weight has met the target. Total pieces include good pieces defects, which consist of underweight and overweight products. OEE categorizes the significant losses from poor performance and explores the manufacturing process's hidden potential. Implementing and improving OEE is also a collaboration between machine operations, maintenance, management, engineering, and planning. The major components of OEE are calculated by the following equation [7]:

$$Availability = \frac{run\ time}{total\ time}$$

$$Performance = \frac{current\ performance}{theoretical\ performance}$$

$$Quality = \frac{good\ pieces}{total\ pieces}$$

These metrics lead to decision-making for equipment management and production planning [9]. The OEE is calculated by the following equation:

$$OEE = Availability \times Performance \times Quality$$

In OEE, the three primary components are performance, quality, and availability. In production, there are a large number of factors associated with production loss. Six big losses are derived from the three OEE metrics, and helps identify and locate the faults during the production line [10]. The Six big losses can be due to equipment failure (availability), set up adjustments (availability), idling and minor stops (performance), reduced speed (performance), process defects (quality), and reduced yield (quality) [1].

SCADA systems are used for condition monitoring. It collects a large quantity of data in real-time. Many researchers have used data mining and machine learning techniques to predict the machines' potential failures. Moreover, it is critical to compare predicted performance patterns against operational data to simulate orientation patterns.

OEE metrics in the SCADA system provide systematic insights into the Manufacturing Execution System (MES).

The detection of downtime events, automatic information and data acquired from the concurrent production line enables the OEE metrics to provide the real picture of the manufacturing process [11].

OEE metrics are an index of the efficiency of the production line. It is commonly used to compare each day's performance and results in an analytical way [12]. Previous studies had demonstrated that the use of SCADA in the food industry, helped to monitor the temperature, pressure, weight in the process, and stirring speed in a mixing machine. The real-time data visualization provided the means for a rapid reaction from the management team to the shopfloor level [3].

Another study had proposed a data-driven approach for optimizing OEE by utilizing discrete event simulation with data analytics tools (Power BI). The identification of six big losses and individual OEE scores of each machine helped them find the areas that needed improvement and identified the factory's bottleneck. However, the limitation of their approach is also obvious since Power BI is not fully capable of transferring big data [6]. Production flow can be varied and complex due to many unplanned downtime events. Monitoring the production line in real-time is essential in improving productivity and identifying issues. In order to have Real-time OEE, research in Peru has designed OEE dashboards in the SCADA system that are connected through PLC, which support real-time measures for each OEE loss and detects the bottleneck of each production line [13]. To improve manufacturing management, one of the major roles of industry 4.0 is to have production analysis to gain insights into machinery status, production plans [2], and help to determine the performance gap from historical IoT data. It helps manufacturers better understand their overall production performance. It is also an essential step in the OEE phase. Daily and weekly performance metrics are formed as a guide to understand how well the real-time was performed.

In this project, we aimed to design a real-time OEE system. The real-time data transfer capabilities in the SCADA system are the primary requirement for this project. The SCADA system must display the OEE metrics in multiple statistical dimensions regarding quality, performance, and availability. In addition, no existing SCADA system has integrated production analysis and OEE real-time monitoring that provides real-time traceability, visibility, and interoperability [2]. Utilizing the information provided by OEE is one of the primary means of pinpointing the manufacturing weaknesses.

We had three months to achieve the core functionalities for this research project. The first month involved project initiation and development environment set up. The second month involved developing in Ignition SCADA, and the third month consisted of reporting, and documentation.

III. OEE ARCHITECTURE

A. Experiment setup

Leaderbrand has organised electricians and automation engineers to set up multiple sensors and wired connections

through Allen Bradley drivers. Sensors were primarily placed in five checkweighers. Ignition SCADA was licensed and responsible for the connection of the OPC server through OPC-UA. Overall, 75 tags and a historical database were prepared for the Tag Historian Module.

Quantitative research has been conducted to develop OEE metrics in the SCADA system. The first initial phase of this project was to identify the problems during the production process. Systematic views of the state-of-the-art solutions helped identify downtime issues which we formulated those findings as our objectives. The second phase is data collection and processing: we used five check weighers to calculate our machine speed, product output, and downtime events. Check weighers can check the weight of the product and record the datetime. We can use the weight to check the quality of machine status and machine speed. It was the best machine option to use in the scope of our study. The last stage was to analyse the data and find the corresponding parameters in terms of the three OEE metrics: quality, performance, and availability. Our proposed real-time OEE process flow as shown in Figure 1.

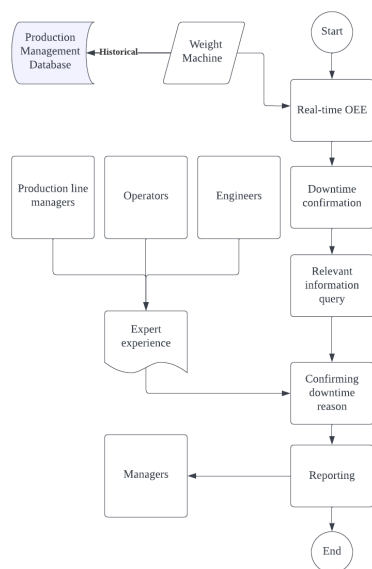


Fig. 1: Real-time OEE process flow

B. Performance analysis for each line

Based on the total bag count from the checkweighers, we can calculate each minute's bag output and create multiple analytical charts that visualise the current production performance and capacity. This step significantly helped LeaderBrand have comprehensive insights into their factory in real-time. Once we calculated the bag per minute speed, we could then categorise and detect the downtime occurrence if the bags per minute output reached 0. This functionality helped us to capture the downtime and loss per minute in each product in each line and stored them in the database for further experimentation.

Bags per minute can also be extended to use prediction, we can use the machine's performance to calculate production finish time and output at the end of the day. Calculating the performance was a crucial step for this project, where downtime can be categorised as availability, bags per minute can be calculated as performance and defects from the check weigher (underweight or overweight) can be used as a metric for quality control. Based on that, we are able to calculate the OEE in terms of each product, each line, and the whole day in real-time. As seen in Figure 2 and Figure 3, a comprehensive view is provided to showcase the potential we can achieve in the factory.



Fig. 2: One line OEE overview



Fig. 3: Whole line OEE overview

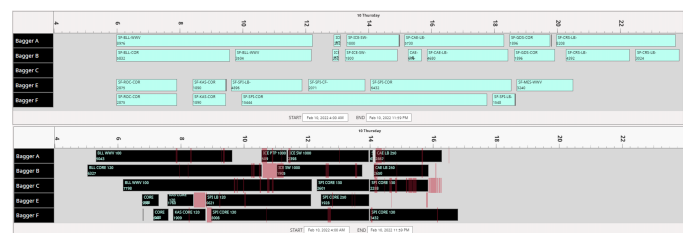


Fig. 4: Production plan against actual operation

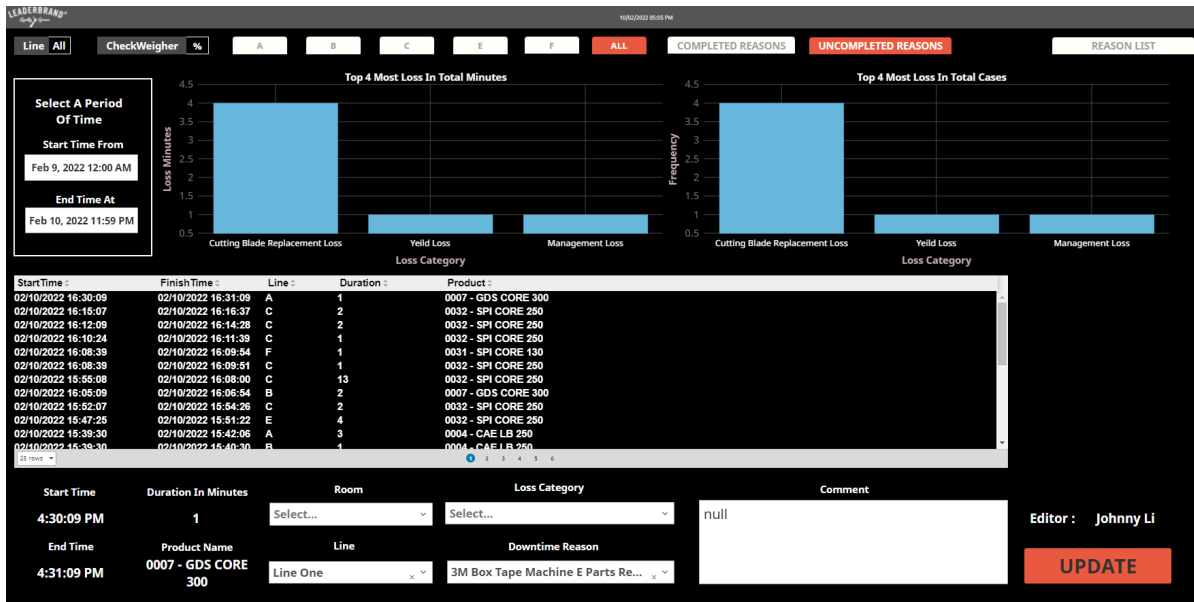


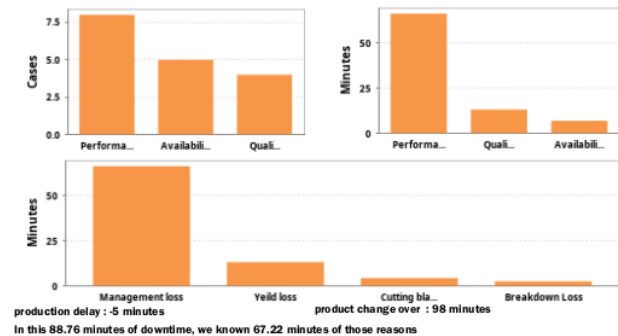
Fig. 5: Downtime capture and category

Production shift DOR report Date 11/02/2022 for 10/02/2022

Line 1			Line 4		
Plan Against Execution					
Line Start Time	Planned 6:00 AM	Actual 5:03 AM	Line Start Time	Planned 6:00 AM	Actual 6:47 AM
Line Finish Time	Planned 11:52 PM	Actual 11:17 PM	Line Finish Time	Planned 11:52 PM	Actual 11:17 PM
Production plan to be executed 57.0 mins earlier			Production plan to be executed 48.0 mins latter		
Change over total min for A	Planned 88 min	Actual 98 min	Change over total min for E	Planned 71 min	Actual 151 min
Change over total min for A	Planned 131 min	Actual 77 min	Change over total min for F	Planned 182 min	Actual 205 min
Available time for A	Planned 985 min	Actual 980 min	Available time for E	Planned 800 min	Actual 839 min
Available time for B	Planned 942 min	Actual 1017 min	Available time for F	Planned 689 min	Actual 757 min
OT mins:	Reason for OT:		OT mins:	Reason for OT:	
High Level Overview					
Line 1 (A, B)		Line 2 (C)		Line 4 (E, F)	
ATP achieved : 81.07 %	Overall RFT : 52598	ATP achieved : <N/A> %	Overall RFT : 15645	ATP achieved : 85.81 %	Overall RFT : 3729
Planned total bag	43751	Planned total bag	0	Planned total bag	52898
Actual total bag	42885	Actual total bag	15645	Actual total bag	38729
Proper total bag	42885	Proper total bag	15302	Proper total bag	37097
Actual total weight	14709.37 Kg	Actual total weight	2119.5 Kg	Actual total weight	6325.95 Kg
Proper total weight	14386.94 Kg	Proper total weight	2058.73 Kg	Proper total weight	6057.26 Kg
Attendance, Labour & Downtime					
Line 1		Line 4			
Attendance % :	Labour Cost :	Attendance % :		Labour Cost :	
Breakdown mins :	Downtime mins :	Attendance % :		Downtime mins :	
Recovery, Quality and Equipment Management					
Line 1		Line 4			
Rovery % :	Actual	Rovery % :		Actual	
Raw Material quality issues:		Raw Material quality issues:			
Equipment settings to manage quality issues:		Equipment settings to manage quality issues:			
Reason For Plan Shortfalls Or Over Runs					
Safety Incidents And Observation					
Line 1		Line 4			
Total number :		Total number :			
Who:	Incident	Who:		Incident	
Action taken		Action taken			
Who:	Incident	Who:		Incident	
Action taken		Action taken			
Who:	Incident	Who:		Incident	
Action taken		Action taken			

(a) Daily report

Bagger A



Reason	Duration	Comment
Tomra Optical sorter set up	24.04	Tomra set up
Elevator belt set up	12.57	Elevator is broken
Weigh belt set up	10.97	Weigh belt set up took too long
Poor Quality Raw Material (Hard Prep)	7.66	Product quality issues
Damaged product (on the line)	2.86	Product damaged
Swan neck (SORMAQ) breakdown	2.61	Swan neck broken
Separating table set up	1.85	Separating table
Cross feed belt set up	1.6	Cross feed
Product loss on the line	1.6	product dropped
Cross feed belt set up	1.45	Cross feed

(b) Downtime report

Fig. 6: Daily operational report (DOR)

C. Plan against execution

This task is critical to the production management team. Based on the data we received from the PLC device, we have depicted the execution against the plan in a Gantt chart and included the downtime between the execution as shown in Figure 4. This provided the management team with a better understanding of the actual process so that they can develop a better plan to avoid underestimation and overestimation during the planning phase.

D. Downtime categorization

OEE measurement is categorized into Availability, Performance, and Quality within six big losses. We integrated Total Productive Maintenance (TPM), as they utilize sixteen losses, this gave us more detailed causes of downtime; therefore, enabling broader investigation into profound issues within the production line. TPM is not only about maintenance, it also covers all aspects of machine operation, installation, and human variables within an enterprise [14]. These sixteen losses consist of: breakdown loss; set up & adjustment loss; cutting blade replacement loss; start-up loss; minor stoppage & idling loss; speed reduction loss; defect&rework loss; shutdown loss; management loss; motion loss; line organisation loss; distribution loss; measuring & adjustment loss; yield loss; energy loss; dig, jig & tool loss.

As a result, we listed our downtime reasons according to the factory assets number as shown in Figure 5 and made a automated report for downtime reason categorization as shown in Figure 6a and Figure 6b. A total of 356 reasons have been listed.

IV. DISCUSSION

Leaderbrand acknowledged the need to develop a OEE SCADA system to provide reliable historical and real-time vital data. Required functionalities also include real-time monitoring of product quality, loss of product through overweight measuring, and raw material consumption. Another requirement was to have responsive design in all other web pages to ensure the client version will be suitable in different devices. The key functionalities of the developed system has been categorized into these features: This system can analyze real-time and historical data and present them in OEE metrics. Downtime events can be recorded and classified for future use. It provided a comprehensive insight of each production run and forecasts its finish time and minutes behind. And also provided production analysis of raw material consumption, quality control, and machinery speed. Compared to the state-of-the-art solutions, there was no similar system like this as shown in Table I.

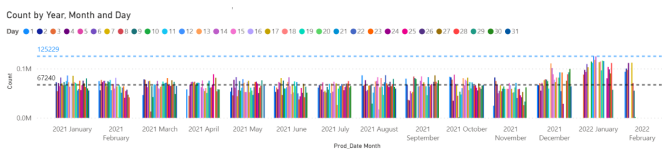


Fig. 7: Production output in 2021 and early 2022

After using the OEE system, we noticed that our most productive time is from 6 a.m. to 10 p.m., Monday through Friday. Previously we ran the production line 20 hours per day, Monday through Saturday. Therefore the management team quickly adjusted the production plan. With production shift adjustment, our OEE project had delivered positive results and benefits, as confirmed by the Leaderbrand case study. Overall capacity was significantly increased above 20% since the project initiation in December 2021, as shown in Figure 7. It helped LeaderBrand drive its production output by providing a comprehensive and straightforward understanding of quality, availability, performance, and improved efficiency and operational cost. This system enables manufacturers to have an intelligent control room of OEE metrics. It helped management teams discover hidden production losses and transforms data into accessible information for the production team, as shown in Figure 8.

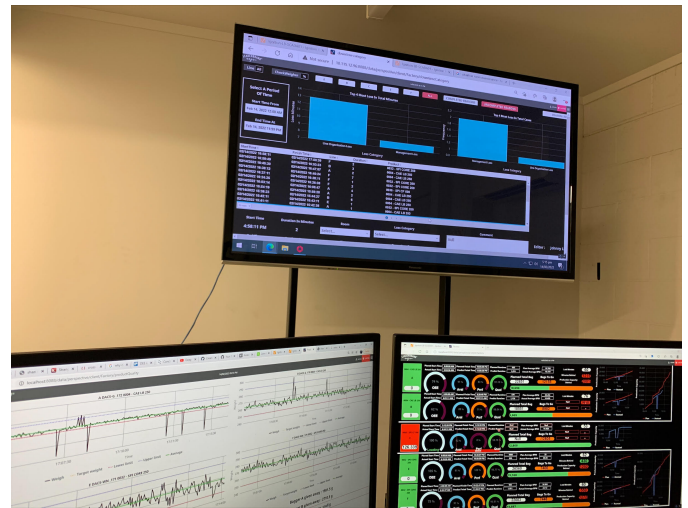


Fig. 8: OEE metrics intelligent control room

V. CONCLUSION AND FURTHER DEVELOPMENT

In terms of the proposed research incentives, we successfully managed to visualize OEE metrics in real-time dashboards through our initial objectives. (Research question 1) we investigated and gathered multiple state-of-the-art solutions as an inspiration and foundation for our project. (Research question 2) In order to provide a high level production analysis and record the case study company's downtime, we needed to combine it with OEE metrics, to enhance the traceability, product quality and production capacity. (Research question 3) As intended, our proposed OEE system provided a comprehensive view of the manufacturing production line regarding performance, quality, and availability. We captured and identified the downtime reasons that caused production loss. We helped visualize actual results against the production plan. We helped automate Leaderbrand's Daily Operational Report as shown in Figure 6a and Figure 6b. However, this project can be extended in

TABLE I: Comparison of state of the art solution

	Real-time Capability	OEE Measurement	Downtime Reason Categorization	Downtime Alarm	Production Analysis
Analysis of total productive maintenancem(TPM) implementation using overall equipment effectiveness (OEE) and six big losses: A case study [14]		x	x		
Remote Supervision of Production Processes in the Food Industry [3]	x	x			
Energy consumption analysis in wastewater treatment plants using simulation and SCADA system: Case study in northern Taiwan [4]	x				x
Design of an Automated Management System to Improve Productivity in a Manufacturing Line Based on Industry 4.0 [13]	x	x	x		
Combining Simulation and Data Analytics for OEE Improvement [6]		x			x

several directions to further improve production capacity at Leaderbrand or other such companies:

- **Machine learning technique:** There is vast potential for LeaderBrand to better control their machine set-up and conditions by analyzing and comparing the product's weight. Additional data from other machines would provide vital values for the analysis and better insights into the relationship between different parameters. With the provided correct machine set-up parameters, we can produce better production output.
- **Financial loss analysis during the manufacturing process:** It is critical to let the management team understand how much money has been lost during manufacturing. It gives us guidance on how Leaderbrand's machines were performed and how much raw materials were given away. This feature enables us to improve production capacity and reduce costs. Currently, this feature is still in the development phase. Eventually, our goal is to implement a machine learning model to learn the pattern of the machine performance, return the optimized parameters, and generate the best setup status for the machine. Therefore, increasing its production output and quality.
- **Downtime reason classification:** Another future work is to use classification for downtime reasons. Since we designed an interface to fill each downtime's reason on the OEE system, we can use that data to merge more machinery parameters into the dataset and use classification technique to identify each downtime duration's likelihood reason automatically.

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